# A cube counterexample to the complementation-set correspondence

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**Task 1 (original problem).** MathDB 380333 records Genest's Conjecture 1.7.17 [1, p. 41]: for a bicolored graph without isolated vertices or twins, its parity class is in bijection with its complementation sets. We disprove the conjecture, including for a graph with its natural coloring.

**Theorem 1.** *Let $G$ be the three-dimensional cube, all of whose vertices are black. Then $G$ is connected, has no twins or isolated vertices, has exactly* 56 *complementation sets, and has exactly* 28 *graphs in its parity class. Consequently no bijection in the conjecture exists.*

**Definitions from the source.** Local complementation at $u$ toggles the edges between distinct neighbors of $u$. If $u$ is white, it also reverses its neighbors' colors; if $u$ is black, all colors stay unchanged. Allowed parity steps are a white singleton $u$, or $[uv] = uvu$ for adjacent black vertices $u, v$. A parity word is any finite sequence of allowed steps, including the empty word. The parity class $[G]$ consists of the resulting colored graphs on the original vertex set. A reduced word uses disjoint singleton or pair supports; its support is a *complementation set*. The result for support $S$ is denoted $G^S$ and is independent of the reduced word. These are Definitions 1.6.1–1.6.4 and Theorem 1.6.3 in [1]. Twins mean distinct vertices with equal open neighborhoods or equal closed neighborhoods. The natural coloring makes odd degrees black and even degrees white.

Starting with an all-black graph, every parity step is an adjacent black pair, and every resulting graph is all black. We work entirely in this case. All matrices below are over $\mathbb{F}_2$.

**Lemma 1.** *Let $A$ be the adjacency matrix of an all-black graph. Its complementation sets are exactly the sets $S$ for which the principal submatrix $A[S]$ is nonsingular, with the empty matrix regarded as nonsingular. Every graph in its parity class equals $G^S$ for such a set.*

*Proof.* For a nonsingular principal block $P = A[S]$, write

$$A = \begin{pmatrix} P & Q \\ Q^t & R \end{pmatrix}.$$

Solving the first block of $y = Ax$ for $x_S$ gives

$$\begin{pmatrix} x_S \\ y_{S^c} \end{pmatrix} = \underbrace{\begin{pmatrix} P^{-1} & P^{-1}Q \\ Q^tP^{-1} & R + Q^tP^{-1}Q \end{pmatrix}}_{A*S} \begin{pmatrix} y_S \\ x_{S^c} \end{pmatrix}. \tag{1}$$

This defines the principal pivot $A * S$.

For an adjacent black pair $u, v$, its block is $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) = P^{-1}$. If its outside incidence rows are $c^t, d^t$, formula (1) swaps these rows and adds $cd^t + dc^t$ to the outside block. The source's three local

complementations $uvu$ do exactly this: outside adjacencies $(c_i, d_i)$ become $(d_i, c_i)$, and the toggle between outside vertices $i, j$ is

$$c_i c_j + (c_i + d_i)(c_j + d_j) + d_i d_j = c_i d_j + d_i c_j.$$

All colors stay black.

For completeness, let

$$L_A = \{(x, Ax) : x \in \mathbb{F}_2^n\},$$

and let $\sigma_S$ interchange $x_i$ and $y_i$ for $i \in S$. These swaps commute and satisfy $\sigma_S \sigma_T = \sigma_{S \triangle T}$. Writing $D_S$ for the diagonal indicator of $S$, the first coordinate of $\sigma_S(x, Ax)$ is

$$C_S x, \qquad C_S = I + D_S + D_S A.$$

Its rows outside $S$ are identity rows and its rows inside $S$ are rows of $A$. Therefore $\det C_S = \det A[S]$. It follows that $\sigma_S L_A$ is the graph of a matrix precisely when $A[S]$ is nonsingular; that matrix is $A * S$.

Any parity word is a sequence of pair swaps. Its final subspace is $\sigma_S L_A$, where $S$ is the symmetric difference of all its pair supports. The final graph is a matrix graph, so $A[S]$ is nonsingular and the final matrix is $A * S$.

Conversely, if $A[S]$ is nonsingular and nonempty, it is alternating: it is symmetric with zero diagonal. It has a nonzero off-diagonal entry, hence an allowed black pair. Its Schur complement on the unused part of $S$ is again alternating and nonsingular, by the displayed formula and determinant factorization. Repeating gives disjoint pair steps covering $S$. This is a reduced parity word, and (1) shows that it realizes $A * S$. □

**Lemma 2.** *Suppose that $A$ is the adjacency matrix of a connected bipartite graph and $A^2 = I$. For any two complementation sets $S, T$,*

$$G^S = G^T \quad \Longleftrightarrow \quad T = S \text{ or } T = V(G) \setminus S.$$

*In particular, every graph in the parity class has exactly two complementation sets.*

*Proof.* By Lemma 1, equality is equivalent to $\sigma_{S \triangle T} L_A = L_A$. Let $D$ be the diagonal indicator of a set $U$ with $\sigma_U L_A = L_A$. The two coordinates of the swapped vector are

$$x' = (I + D + DA)x, \qquad y' = (D + (I + D)A)x.$$

Because $y' = Ax'$ for every $x$, we obtain

$$D + DA + AD + ADA = 0. \tag{2}$$

For adjacent vertices $i, j$, bipartiteness implies that they have no common neighbor, so $(ADA)_{ij} = 0$. The $(i, j)$ entry of (2) is therefore $d_i + d_j = 0$. Connectivity forces all the $d_i$ to be equal. Thus only $U = \varnothing$ or $U = V(G)$ can fix $L_A$. Both do fix it: the empty swap is the identity, and the full swap gives $L_{A^{-1}} = L_A$ because $A^2 = I$.

Consequently equality of two pivot results holds exactly when $S \triangle T$ is empty or full. Moreover the complement of every feasible $S$ is feasible, since $\sigma_{V \setminus S} L_A = \sigma_S L_A$ is a matrix graph. The two supports are distinct, completing the proof. □

*Proof of Theorem 1.* Label the bipartition as

$$L = \{\ell_1, \ell_2, \ell_3, \ell_4\}, \qquad R = \{r_1, r_2, r_3, r_4\},$$

with $\ell_i$ adjacent to $r_j$ precisely when $i \neq j$. This is the cube: the labels can be taken as

$$(\ell_1, \ell_2, \ell_3, \ell_4) = (000, 011, 101, 110), \qquad (r_1, r_2, r_3, r_4) = (111, 100, 010, 001).$$

Edges then join binary strings differing in one coordinate. The graph is connected and every vertex has degree three. Thus the all-black coloring is its natural coloring.

Two vertices in the same part omit different opposite vertices from their open neighborhoods, so these neighborhoods differ. Open neighborhoods of vertices in different parts are nonempty and lie in disjoint parts, so cannot agree either. Equal closed neighborhoods of distinct vertices would require them to be adjacent. But for an adjacent pair, one closed neighborhood contains one left vertex and the other contains three left vertices. Hence there are no twins.

Let $J_4$ be the all-ones matrix and put $B = I_4 + J_4$. Then

$$A = \begin{pmatrix} 0 & B \\ B & 0 \end{pmatrix}, \qquad J_4^2 = 0, \qquad B^2 = I_4, \qquad A^2 = I_8.$$

It remains to count the nonsingular principal submatrices. For $S = X \cup Y$, with $X \subseteq L$, $Y \subseteq R$, the principal matrix is $\left(\begin{smallmatrix} 0 & B[X,Y] \\ B[X,Y]^t & 0 \end{smallmatrix}\right)$. It can be nonsingular only if $|X| = |Y| = k$, and in that case it is nonsingular exactly when $B[X, Y]$ is. Identify $X, Y$ with their index sets in $\{1, 2, 3, 4\}$. The possibilities are:

| $k$ | condition for nonsingularity | number |
|---|---|---|
| 0 | empty | 1 |
| 1 | $X \neq Y$ | 12 |
| 2 | $\lvert X \cap Y \rvert = 1$ or 2 | $24 + 6 = 30$ |
| 3 | $X \neq Y$ | 12 |
| 4 | $X = Y = \{1, 2, 3, 4\}$ | 1 |

For $k = 2$, disjoint sets give an all-ones matrix of rank one; intersection sizes one and two give, up to row and column permutations, respectively

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

both of determinant one. For $k = 3$, equal sets give $I_3 + J_3$, which kills the all-ones vector, while unequal sets give, up to permutations,

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

of determinant one. The other rows are immediate, using $B^2 = I_4$ for the last row. There are therefore $1 + 12 + 30 + 12 + 1 = 56$ complementation sets. Lemma 2 gives precisely $56/2 = 28$ parity-class graphs. This strict cardinality discrepancy excludes every possible bijection, not only the canonical map $S \mapsto G^S$. □

**An infinite counterfamily.** Every odd-dimensional cube $Q_d$, $d \geq 3$, with its natural all-black coloring has the same two-to-one correspondence. Indeed, write $A = P_1 + \cdots + P_d$, where $P_i$ flips coordinate $i$. The commuting involutions $P_i$ satisfy $A^2 = dI = I$ over $\mathbb{F}_2$. The graph is connected and bipartite. Distinct vertices have at most two common neighbors, whereas every vertex has $d \geq 3$ neighbors, so open twins are impossible. Closed twins are impossible too: they would be adjacent, but an adjacent pair has no common neighbor and degree at least three. Lemma 2 applies.

**Verification and scope.** The exact verifier separately explores unrestricted parity words and reduced words using the source's three local complementations. It checks all 256 supports against independent binary elimination, finds 56 supports and 28 distinct labeled colored graphs, and confirms that every fiber is a complementary pair. For the labels above, one full reduced word returning to the original graph is

$$[\ell_1 r_2][\ell_2 r_1][\ell_3 r_4][\ell_4 r_3].$$

The proof above is independent of this computation. The conjecture fails under its full source definitions, with no transcription repair or missing-hypothesis counterexample involved.

**Task 2 (proposed follow-up).** Determine the number of complementation sets of $Q_d$ for odd $d \geq 5$, and hence the cardinality of its parity class.

**Task 3 (proposed follow-up).** Determine the smallest order of a naturally colored graph without isolated vertices or twins for which the correspondence is not injective.

**Task 4 (proposed follow-up).** Classify the connected bipartite graphs whose binary adjacency matrix satisfies $A^2 = I$, by their degree and common-neighbor constraints.

**Task 5 (proposed follow-up).** For general bicolored graphs, classify the diagonal indicator matrices $D$ satisfying $D + DM + MD + MDM = 0$, where $M$ has diagonal one at white vertices, and determine the possible dimensions of this space for graphs without twins.

## References

[1] F. Genest, *Graphes eulériens et complémentarité locale* (English arXiv title: *Eulerian graphs and local complementation*), Ph.D. thesis, Université de Montréal; arXiv:math/0701421v1 (2007), 103 pages. Definitions 1.6.1–1.6.4, Theorem 1.6.3, and Conjecture 1.7.17. `https://arxiv.org/abs/math/0701421`.